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# Mobile Intelligent System for VARK-Based Student Learning Style Classification Using KNN

**Muhammad Saiful Anwar[1)*], Sri Wulandari[2]**

[1)2)] *Department of Informatics, Faculty of Science and Technology, Universitas Teknologi Yogyakarta, Sleman, Indonesia*

[1)]saian1p4u@gmail.com

[2)]sri.wulandari@staff.uty.ac.id

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***Abstract***

The growing complexity of learning activities in higher education highlights the need for accurate and scalable mechanisms to identify students' learning preferences. Conventional VARK-based assessments, which rely on manual self-report questionnaires, remain limited by subjectivity, low practicality, and the absence of real-time feedback. This study addresses these challenges by developing *Bandela*, a mobile intelligent system that integrates the VARK framework with the K-Nearest Neighbors (KNN) classification algorithm to provide automated learning style identification. Using a Research and Development (R&D) approach, the system was implemented through a three-tier architecture consisting of a Flutter frontend, a Python Flask backend, and a MySQL database. Questionnaire responses collected from students were used as both training and testing datasets for the KNN model, enabling real-time classification across Visual, Auditory, Read/Write, and Kinesthetic categories. Functional evaluation through Blackbox Testing demonstrated that all core features ranging from authentication and questionnaire completion to classification processing, visualization, and community interaction performed reliably and as intended. The findings indicate that Bandela offers an accessible and empirically grounded tool for identifying learning preferences, contributing to more personalized and adaptive learning strategies. This work underscores the practical value of mobile intelligent systems in advancing data-driven personalization within higher education and provides a foundation for future enhancements involving expanded datasets and exploration of additional machine learning techniques.

## I. INTRODUCTION

The effectiveness of the learning process in higher education remains a central determinant of students' academic achievement and long-term intellectual development. At this level, learners encounter increasingly complex academic tasks that require not only cognitive competence but also the ability to adopt learning strategies aligned with their individual characteristics. One essential factor influencing learning effectiveness is the alignment between instructional methods and students' distinct learning preferences. Each student processes, absorbs, and organizes information differently patterns that are collectively conceptualized as learning styles [1]. Prior research consistently highlights that a deep understanding of learning styles can lead to improved academic performance, heightened motivation, and enhanced learner satisfaction in university environments [2], [3]. Despite these benefits, many students remain unaware of their dominant learning style, resulting in the adoption of study approaches that are subjective, unfocused, and inefficient. Such misalignment often leads to difficulties in comprehending course materials, reduced engagement, and ultimately suboptimal academic outcomes [4]. The rapid digital transformation in educational settings further underscores the need to equip students with personalized, data-driven tools that can align their learning behavior with evidence-based strategies [5].

This challenge becomes more apparent when considering the limitations of traditional learning style identification methods, which commonly rely on paper-based or manual questionnaires. Although widely used, these instruments are often impractical for large-scale deployment, and their results can be influenced by interpretation bias, inconsistent self-reporting, and lack of real-time feedback mechanisms. Meanwhile, data-

* Corresponding author

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driven approaches particularly data mining techniques have shown substantial potential in capturing patterns of learning behavior more objectively. Studies employing clustering algorithms, for example, have demonstrated their ability to group learners with similar behavioral tendencies and cognitive preferences more systematically than manual identification methods [6]. Nevertheless, these approaches are often implemented within research environments rather than integrated into practical, accessible tools that students can use independently. To bridge this gap, a more innovative identification system is needed, one that is not only accurate but also easily accessible, scalable, and capable of supporting personalized and adaptive learning experiences. Adaptive learning itself represents an increasingly prominent pedagogical paradigm, wherein teaching strategies, learning pathways, and instructional materials are dynamically adjusted to suit the unique needs of individual learners [7].

In parallel with these pedagogical developments, technological advancements have fundamentally reshaped the landscape of higher education. The integration of digital systems into learning environments has enhanced the flexibility, accessibility, and interactivity of educational processes, allowing students to engage with learning materials anytime and anywhere. Empirical evidence demonstrates that technology adoption in education can improve the overall learning experience, strengthen student engagement, facilitate personalized resource allocation, and promote self-regulated learning behaviors [8]. Among the various technological modalities available, mobile applications have emerged as particularly promising tools due to their widespread use among students, ubiquity across socioeconomic backgrounds, and ability to deliver personalized learning experiences in real time [9]. When combined with artificial intelligence (AI) techniques, mobile platforms can transform traditional educational practices into intelligent learning ecosystems. Machine learning algorithms, such as the K-Nearest Neighbors (KNN) classifier, offer robust capabilities for analyzing learner data and producing accurate, pattern-based predictions. Through systematic comparison of learner characteristics and response patterns, KNN can generate real-time feedback on learning tendencies, thereby providing an evidence-based foundation for personalized learning recommendations [10].

The successful implementation of technological solutions in educational contexts, however, is contingent not only on algorithmic accuracy but also on system reliability, functional stability, and user-centric design principles. For this reason, the present study concentrates on the design and development of *Bandela*, a mobile-based system that integrates the VARK learning style framework with the KNN classification algorithm to generate accurate learning style profiles in real time [11]. Beyond its computational components, *Bandela* emphasizes interface design quality as a central factor influencing usability and user satisfaction. Previous studies have shown that visual design elements such as typography, color harmony, and letterform composition play a critical role not only in conveying information but also in shaping users' cognitive and emotional responses to digital interfaces [12], [13]. A well-designed interface can significantly enhance ease of navigation, reduce cognitive load, and facilitate intuitive interaction, all of which are essential for maintaining user engagement in educational applications.

Furthermore, *Bandela* incorporates gamification strategies to increase student motivation through reward elements, progress indicators, and interactive feedback loops. Existing literature confirms that gamification can enhance student engagement, intrinsic motivation, and learning persistence, especially when integrated into digital learning systems [14]. Complementing this approach, cognitive ergonomics principles were employed to create an interface that aligns with human perceptual and cognitive capabilities, thereby promoting efficiency, reducing task complexity, and improving overall user experience [15]. The combination of adaptive learning concepts, machine learning-based classification, user-centered interface design, and motivational gamification positions *Bandela* as a comprehensive and innovative system tailored to the learning needs of higher education students.

Through these contributions, this research aims not only to produce a functional mobile application prototype but also to establish a foundational step toward the broader implementation of adaptive learning ecosystems in higher education. By providing students with accurate, accessible, and personalized insights into their learning tendencies, *Bandela* offers the potential to enhance study effectiveness, support educators in designing tailored instructional strategies, and strengthen the overall quality of digital learning environments.

## II. RELATED WORKS/LITERATURE REVIEW

Current research on learning style identification presents a wide range of methodological approaches and technological innovations. Understanding these approaches is essential to position the contribution of the present study within the broader development of educational technology. Several studies have investigated the use of the VARK model in educational settings, demonstrating that understanding individual learning preferences can enhance learning effectiveness and student performance [1]. Although this work provides a strong theoretical foundation, it does not extend to the development of automated systems for learning style identification.

A related study proposed an adaptive learning model that integrates VARK-based questionnaires with Learning Analytics to predict learning styles and provide real-time intervention [7]. However, the resulting model remains conceptual, relying on literature analysis and flowchart-based frameworks, with the primary focus on providing insights for educators rather than building a practical tool for students. In contrast, the present research offers a functional mobile application that employs the K-Nearest Neighbors (KNN) algorithm for direct learning style classification.

Another line of research introduced an expert system for diagnosing learning styles using the Forward Chaining method [16]. While effective, this rule-based mechanism requires manually crafted rules, limiting scalability and adaptability. The present system improves upon this by using a data-driven algorithm capable of automatic classification based on empirical response patterns.

Descriptive research examining learning styles in higher education has identified diverse learning tendencies, such as competitive, avoiding, and independent orientations [17]. Although these studies contribute valuable insights into student characteristics, they do not incorporate system implementation or automated feedback capabilities. Conversely, the current study provides real-time output through a mobile interface.

In the context of technological adoption, research on mobile learning applications highlights the high accessibility, flexibility, and user acceptance of mobile platforms in higher education [9]. These findings reinforce the selection of a mobile medium for the present system. Studies focusing on classification algorithms have also demonstrated the effectiveness of KNN for pattern recognition and classification tasks involving small to medium datasets, supporting its suitability for learning style identification [10].

From a user interface design perspective, investigations into typography and UI/UX elements reveal their substantial influence on user perception, cognitive processing, and overall usability [12], [13]. These insights inform the design choices in the present system, where visual elements are used to enhance both aesthetics and user comprehension. Gamification research provides evidence that game-based elements can significantly increase user motivation and engagement in learning environments [14]. This supports the potential integration of motivational features in future development stages of the system. Meanwhile, system validation studies employing Black Box testing emphasize the need for functional accuracy and performance stability in educational applications [18], a principle adopted in evaluating the prototype.

Usability research using models such as Nielsen's heuristics and the System Usability Scale (SUS) underscores the importance of user-centered evaluation to ensure an optimal user experience [19]. Complementary to this, design frameworks adopting User Centered Design (UCD) highlight the necessity of aligning system features with user needs and expectations [20].

Overall, existing research indicates substantial progress in understanding learning styles and developing related frameworks; however, a significant gap remains in integrating the VARK model with machine learning specifically KNN within a mobile-based platform that is accessible, user-friendly, and capable of generating personalized results in real time. The present research addresses this gap by introducing *Bandela*, an intelligent mobile-based system designed to facilitate adaptive learning in higher education environments.

## III. METHODS

This research employs a Research and Development (R&D) approach to produce a functional mobile application. The R&D method involves systematic stages that guide the process of developing and refining a product. It also serves as a bridge between basic research and applied research, ensuring that theoretical concepts can be transformed into practical solutions [21].

### A. Research and Development (R&D) Stages

The implementation of the Research and Development (R&D) method in this study follows a structured and systematic framework. This framework is organized into several essential stages that guide the development process from initial problem identification to product refinement. An overview of these stages is summarized in Table 1 to illustrate the progression of the research.

TABLE 1.
RESEARCH AND DEVELOPMENT STAGE

| R&D Stage | Implementation in Research |
|---|---|
| Potential and Problem Identification | Observing problems that students have difficulty understanding their own learning styles, and identification methods are still subjective (manual). This stage was conducted through literature study and observation of students and similar applications. |
| Data Collection | Collecting primary data (through digital questionnaires) and secondary data (through literature and observation). The aim is to obtain answer patterns that become the basis of training data for the KNN algorithm. |
| Product Design / System Design | Designing three-tier system architecture (Flutter frontend, Flask backend, MySQL database), and designing login flow, VARK model-based learning style questionnaire completion, and classification. At this stage, UML (Use Case and System Flowchart) is also compiled. |
| Product Development / Implementation | Developing Bandela application prototype based on the design that has been prepared. Implementation is done using Flutter for frontend and Flask (Python) for backend. The KNN algorithm is applied for learning style classification. |
| Product Testing (Testing and Validation) | Conducting Blackbox Testing on all main features (registration, login, questionnaire, identification results, community, etc.) to verify system conformity with specifications. |
| Product Evaluation and Refinement | Evaluating test results and classification result validity. Based on feedback, refinements are made so that the application is more stable and results are more accurate. |

## B. Data Collection

Research data in this study was collected from two main sources, namely primary and secondary data. Primary data was obtained through the distribution of digital questionnaires based on the VARK model to students, which served as both training and testing datasets for the K-Nearest Neighbors (KNN) classification algorithm. Meanwhile, secondary data was gathered from literature studies, scientific journals, and observations of similar applications to deepen understanding of learning style identification and machine learning–based classification models. The VARK questionnaire used in this study has been widely validated in previous research and is recognized for its reliability in identifying student learning preferences The VARK questionnaire has been widely validated and used in various studies to identify student learning styles [6] [3]. After the data collection phase was completed, the questionnaire responses were compiled and summarized, as presented in Table 2, forming the basis for subsequent analysis of student learning style tendencies.

TABLE 2.
STUDENT LEARNING STYLE QUESTIONNAIRE DATA

| Visual | Auditory | Read/write | Kinesthetic | Dominant Style |
|---|---|---|---|---|
| 5 | 7 | 7 | 2 | Auditory |
| 2 | 3 | 0 | 11 | Kinesthetic |
| 2 | 4 | 8 | 2 | Read/write |
| 12 | 7 | 3 | 10 | Visual |
| 4 | 10 | 5 | 6 | Auditory |
| 1 | 9 | 14 | 1 | Read/write |
| 10 | 9 | 5 | 3 | Visual |

## C. System Architecture

This system is designed using a three-tier architecture consisting of user interface (frontend), business logic (backend), and data storage (database) layers. This three-tier structure was chosen to improve scalability, reliability, and ease of long-term system maintenance.

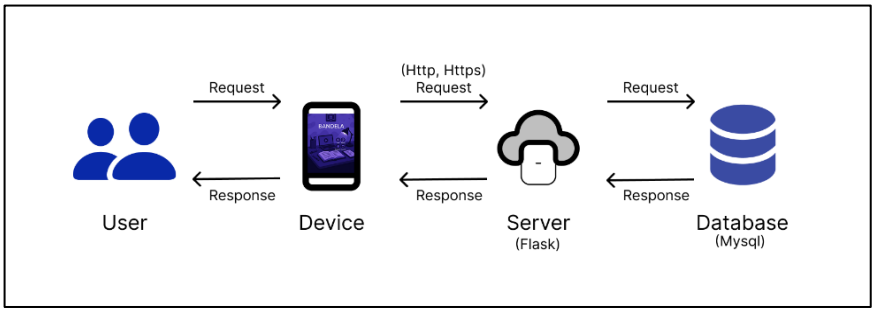



Fig. 1 Three-tier system architecture of Bandela application

Figure 1 illustrates the request flow from users through devices to the server (Flask), then interacting with the database (MySQL) to process data and send responses back.

## D. K-Nearest Neighbors (KNN) Classification Algorithm

Learning style classification is done automatically using the K-Nearest Neighbors (KNN) algorithm. KNN is a supervised learning method that works by comparing the distance between test data and a number of K nearest data in the training data set [10]. Class prediction is determined based on the majority category of the K nearest neighbors. The distance between data is calculated using the Euclidean Distance metric as formulated in the following formula (1) :

$$d(A, B) = \sqrt{\Sigma(a_i - b_i)^2} \quad (1)$$

Where d(A, B) is the distance between new data A and training data B, and $a_i$ and $b_i$ are the i-th features of each data. This method was chosen because of its simplicity, high accuracy for small to medium datasets, and its ability to adapt classification results based on actual data patterns without complex model training [10].

The KNN algorithm is proven effective for pattern-based classification with good accuracy for small to medium datasets [10]. Recent studies have demonstrated that classification models based on machine learning, including KNN, can achieve high accuracy rates in educational contexts. For instance, research on classification models for student assistance programs achieved accuracy rates exceeding 90% when properly trained with relevant datasets [22]. This supports the selection of KNN as the classification algorithm in this research, example implementation:

A_new = [Visual = 8, Auditory = 5, Read/Write = 5, Kinesthetic = 6]
$A_1$ = [5, 7, 7, 2] with label Auditory
$A_2$ = [2, 3, 0, 11] with label Kinesthetic
$A_3$ = [2, 4, 8, 2] with label Read/Write
$A_4$ = [12, 7, 3, 10] with label Visual

$A_5$ = [4, 10, 5, 6] with label Auditory

Calculate distance from A_new to each training data:
Distance to $A_1$ (Auditory label)
$d(A_new, A_1) = \sqrt{[(8-5)^2 + (5-7)^2 + (5-7)^2 + (6-2)^2]} = \sqrt{(9 + 4 + 4 + 16)} = \sqrt{33} = 5.744$
Distance to $A_2$ (Kinesthetic label)
$d(A_new, A_2) = \sqrt{[(8-2)^2 + (5-3)^2 + (5-0)^2 + (6-11)^2]} = \sqrt{(36 + 4 + 25 + 25)} = \sqrt{90} = 9.486$
Distance to $A_3$ (Read/Write label)
$d(A_new, A_3) = \sqrt{[(8-2)^2 + (5-4)^2 + (5-8)^2 + (6-2)^2]} = \sqrt{(36 + 1 + 9 + 16)} = \sqrt{62} = 7.874$
Distance to $A_4$ (Visual label)
$d(A_new, A_4) = \sqrt{[(8-12)^2 + (5-7)^2 + (5-3)^2 + (6-10)^2]} = \sqrt{(16 + 4 + 4 + 16)} = \sqrt{40} = 6.324$
Distance to $A_5$ (Auditory label)
$d(A_new, A_5) = \sqrt{[(8-4)^2 + (5-10)^2 + (5-5)^2 + (6-6)^2]} = \sqrt{(16 + 25 + 0 + 0)} = \sqrt{41} = 6.403$
Sort distance calculation results from smallest to largest and perform voting based on K = 3:
$A_1$ (Auditory label; distance = 5.744)
$A_4$ (Visual label; distance = 6.324)
$A_5$ (Auditory label; distance = 6.403)
Using K = 3, looking at the 3 nearest neighbors ($A_1$, $A_4$, $A_5$) with labels 2 Auditory and 1 Visual. Prediction result: Auditory (has the most votes). Therefore, the KNN model will predict that the learning style for new data (A_new) is Auditory.

### E. UML Diagrams

System modeling in this research uses Unified Modeling Language (UML) to describe the relationship between users and the system as well as the process flow that occurs within the application [23]. UML was chosen because it can provide a structured visual representation of the system design before the implementation stage is carried out. In this research, modeling is focused on two main diagrams, namely Use Case Diagram and System Flowchart.

1) Use Case Diagram

The use case diagram is compiled to describe the relationship between actors and the system based on functional requirements that have been identified in the design stage.

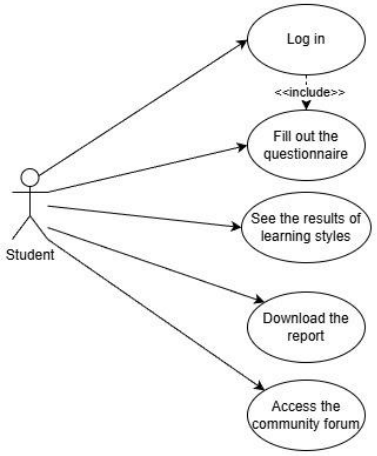



Fig. 2 Use case diagram showing student interaction with the Bandela system

Figure 2 depicts the interaction between the main actor, namely Students, with the Bandela system. Actors can perform login processes, complete VARK model-based learning style questionnaires, view learning style identification results, download result reports, and access community forums. The «include» relationship between Complete Questionnaire and Login shows that the questionnaire completion process can only be done after the user successfully logs into the system.

2) System Flowchart

Figure 3 illustrates the sequential flow of the Bandela system, beginning with the user authentication process. Users are first required to enter their username and password, after which the system validates the submitted credentials. If the login information is correct, users are directed to the dashboard and main navigation menus where they can access various features. One of the key features available at this stage is the learning style questionnaire, which users must select before proceeding to the next step.

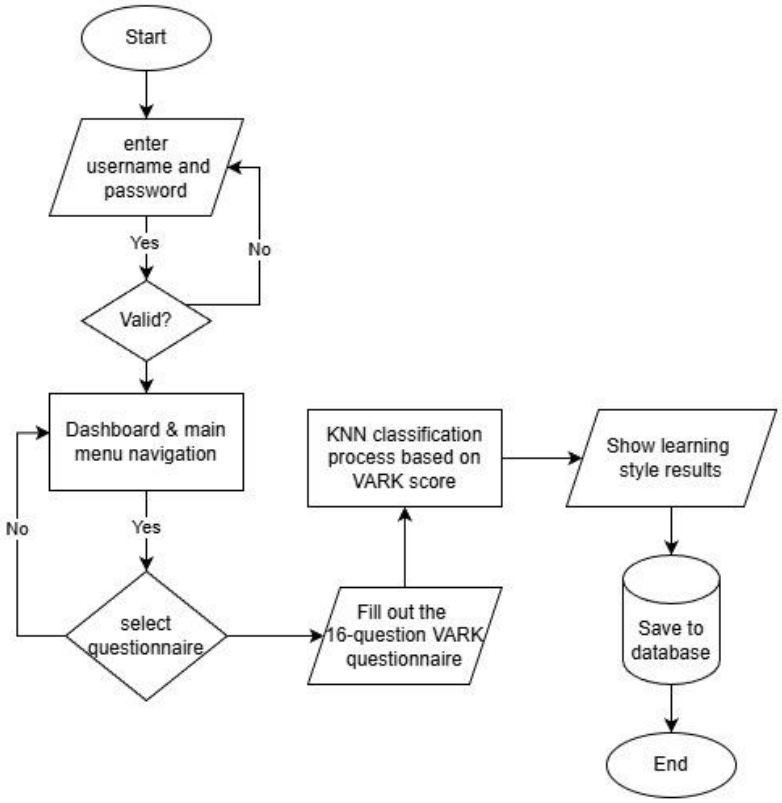



Fig. 3 System flowchart of learning style identification process

Once the questionnaire is selected, users are prompted to complete 16 questions based on the VARK learning style model. After all responses are submitted, the system performs an internal calculation of VARK category scores and applies the K-Nearest Neighbors (KNN) algorithm to classify the dominant learning style. The resulting identification is then displayed on the user interface, providing immediate feedback to the user. Finally, the classification output along with the questionnaire data is saved automatically into the database, marking the completion of the entire process.

## IV. RESULTS

The results of this research are a functional "*Bandela*" mobile application prototype that has undergone testing to validate its performance.

1) Home and Community Display

The application interface has several pages accessed through the bottom navigation bar. The Home page displays the homepage with learning style information and access to the application's main features. Meanwhile, the Community page functions as a discussion forum that allows users to interact, share experiences, and exchange learning-related information.

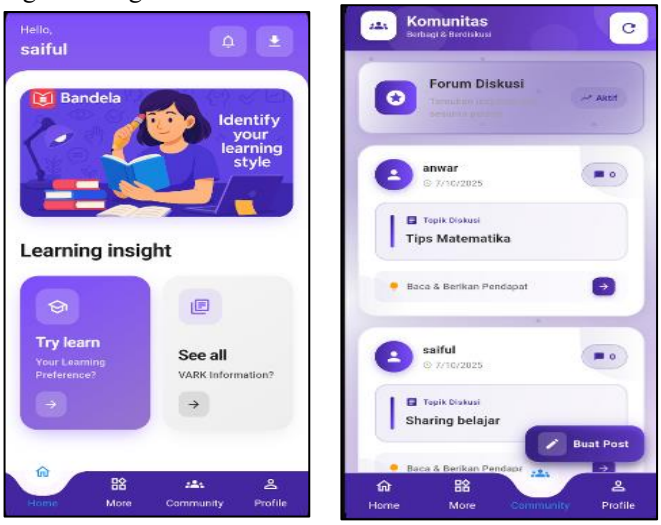


Fig. 4 Home page displaying learning style information and Community forum page for user interaction and discussion

2) Questionnaire Completion and Identification Results Display

The application's main feature includes a learning style identification flow that starts from the questionnaire completion stage, followed by displaying identification results presented visually in the form of pie charts, so users can understand their learning style tendencies more easily and informatively. Through data visualization, information from analysis results can be explored, understood, and communicated more clearly in the form of attractive graphic displays [24].

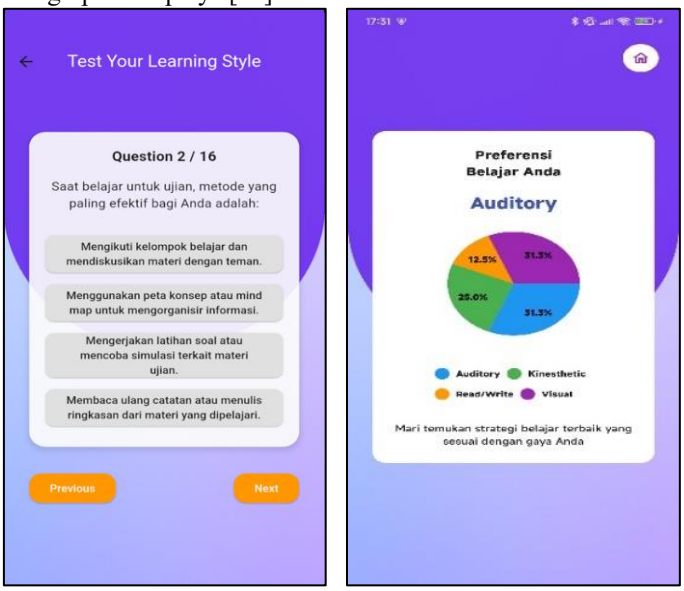



Fig. 5 Questionnaire completion interface and learning style identification results with pie chart visualization

3) Functional Testing

System functionality testing has been carried out using the Blackbox Testing approach to verify the conformity of all main features of the Bandela application [18]. The testing process was conducted through simulation of critical features, including registration, login, forgot password, edit profile, questionnaire completion, learning style classification, community features, and result download. Each function is evaluated based on conformity between actual output and expected output. Table 3 presents a summary of functionality test results for each main feature.

TABLE 3
FUNCTIONALITY TEST RESULTS

| No | Feature | Input | Expected Output | Status |
|---|---|---|---|---|
| 1 | Register | Username/email, password | Account successfully created | Success |
| 2 | Login | Email, password | User successfully logged in | Success |
| 3 | Forgot Password | Email, token, password | New password successfully saved | Success |
| 4 | Edit Profile | Username, email | User successfully changed username & email | Success |
| 5 | Questionnaire Completion | Complete answers to all questions | Successfully sent answers | Success |
| 6 | Identification Results | Questionnaire answers, score calculation, scores sent as KNN model | Receive learning style recommendation results | Success |
| 7 | Download Test Results | Click download learning style feature | Save results in document | Success |
| 8 | Community | Post title, post content | Post successfully created | Success |

Table 3 summarizes the results of the functional testing performed using the Blackbox Testing approach, which evaluates system behavior solely based on input–output conformity without inspecting internal code execution. The tests show that all major features of the Bandela application including registration, login, password recovery, profile editing, questionnaire completion, learning style classification, result downloading, and community interaction operate according to the predefined requirements. Each feature consistently produced outputs that matched the expected results, indicating that the system logic, data flow, and user interface components function reliably. Overall, these outcomes confirm that the application is capable of supporting user interactions smoothly and can be deployed as a stable tool for identifying learning styles and facilitating user engagement.

## V. DISCUSSION

The results of this study demonstrate that the integration of the VARK framework with the K-Nearest Neighbors (KNN) algorithm within a mobile-based architecture offers a practical and effective approach for producing real-time learning style classifications. Compared to earlier studies that primarily proposed conceptual adaptive learning models or descriptive analyses of student learning tendencies [7],[17], the present work provides a fully operational system accessible directly by students. This distinction positions *Bandela* not merely as a theoretical contribution but as an applied technological intervention capable of addressing the limitations of manual VARK assessments, such as subjectivity and the absence of immediate feedback. In addition, the three-tier architecture adopted in this research ensures stable data communication and functional reliability, aligning with findings from prior work on mobile-based educational systems emphasizing the importance of system performance and user interaction quality [9].

Despite these strengths, several methodological considerations warrant further reflection. First, the classification results rely on self-reported questionnaire data, a common constraint in learning style research that may introduce response bias and affect model precision. While KNN is suitable for small- to medium-sized datasets [10], the dataset used in this study remains limited in size, which may constrain the generalizability of the findings. Future work would benefit from expanding the dataset across different faculties or institutions to enhance the robustness of the classifier. Furthermore, potential class imbalance within VARK categories should be examined, as disproportionate representation may influence voting outcomes in KNN and reduce the model's sensitivity to minority learning style groups. As suggested by prior studies on educational data classification [22], the incorporation of cross-validation or parameter tuning techniques could further strengthen the reliability of the model.

The comparative evaluation with related works also highlights the unique contribution of *Bandela* in its integration of machine learning, mobile interaction, and user-centered design. While expert system approaches using rule-based mechanisms such as Forward Chaining have been applied to diagnose learning styles [16], these systems often lack scalability because their performance is constrained by predefined rules that may not adapt well to evolving student behavior. In contrast, the data-driven approach used in this study allows the classification model to reflect patterns in real user responses, offering a more flexible and empirical foundation for personalization. Moreover, the incorporation of UI/UX considerations and community features aligns with broader findings on the importance of usability, motivation, and engagement in digital learning environments [25]. These design choices strengthen the value of *Bandela* as a holistic learning support tool rather than merely a classification engine.

Looking ahead, the findings of this study provide a promising foundation for the development of more advanced adaptive learning ecosystems. The system could be enhanced by integrating behavioral analytics, such as user interaction patterns or learning activity logs, to generate dynamic recommendations beyond static VARK profiles. Additionally, the scalability of the system could be further examined by evaluating server load, response time, and model performance in larger deployments, particularly if the system is intended for institutional use. The incorporation of alternative machine learning algorithms such as Support Vector Machines, Random Forests, or neural network models may also offer improved accuracy and deeper insights into complex learning behavior patterns. These extensions would align with global trends in intelligent educational systems and strengthen the system's contribution to data-driven personalization in higher education.

Overall, the revised discussion maintains that *Bandela* meaningfully advances the field by transforming validated psychometric instruments and machine learning methods into a practical mobile solution capable of supporting more personalized, efficient, and accessible learning experiences. By acknowledging its limitations and outlining concrete pathways for enhancement, this study contributes both an operational system and a roadmap for future research on intelligent, mobile-based learning analytics.

## VI. CONCLUSIONS

This study demonstrates that the integration of VARK-based digital questionnaires with the K-Nearest Neighbors (KNN) algorithm within the Bandela mobile application provides a reliable and accessible mechanism for identifying student learning preferences in real time. Through a three-tier architecture and comprehensive functional testing, the system successfully delivers stable performance across its core features, including authentication, questionnaire processing, classification output, and visualization. These results underscore the system's practical value for empowering students to better understand their cognitive tendencies and adopt more targeted study strategies, while also offering educators actionable insights that can support the design of adaptive instructional materials. In this regard, the study contributes a tangible, data-driven tool that advances personalization efforts within higher education settings.

Building upon these findings, the study also highlights opportunities for further refinement and expansion. Future research may explore the incorporation of larger and more diverse datasets to enhance the generalizability and robustness of the classification model, as well as examine alternative machine learning approaches such as

Support Vector Machines or ensemble-based classifiers to compare performance across algorithms. Additional enhancements, including cross-institutional validation, the integration of behavioral learning analytics, and potential multilingual or cross-cultural adaptations, could further strengthen the system's applicability in broader educational contexts. These directions not only address the limitations of the current version but also position Bandela as a promising platform for advancing personalized learning in low-resource and mobile-first academic environments.

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